

Auto loan prediction

This is a Markdown cell

Yeshhhh

wooo

A data.frame: 40 × 2

Variable	Description
<chr>	<chr>
ID	Client Loan application ID
Client_Income	Client Income in \$
Car_Owned	Any Car owned by client before applying for the loan for another car (0 means No and 1 means otherwise)
Bike_Owned	Any bike owned by client (0 means No and 1 means otherwise)
Active_Loan	Any other active loan at the time of application of loan (0 means No and 1 means otherwise)
House_Own	Any house owned by client (0 means No and 1 means otherwise)
Child_Count	Number of children the client has
Credit_Amount	Credit amount of the loan in \$
Loan_Annuity	Loan annuity in \$
Accompany_Client	Who accompanied the client when client applied for the loan
Client_Income_Type	Clients income type
Client_Education	Highest level of education achieved by client
Client_Marital_Status	Marital status of client (D- Divorced, S- Single, M- Married, W- Widowed)
Client_Gender	Gender of the Client
Loan_Contract_Type	Loan Type (CL- Cash Loan, RL- Revolving Loan)
Client_Housing_Type	Client Housing situation
Population_Region_Relative	Relative population of the region where the client is living. Higher value means the client is living in more populated area
Age_Days	Age of the client at the time of application submission

Employed_Days	Days before the application, the client started earning
Registration_Days	Days before the loan application, the client changed his/her registration
ID_Days	Days before the loan application, the client changed his/her identity document with which the loan was applied
Own_House_Age	Age of Client's house in years
Mobile_Tag	Mobile Number provided by Client (1 means Yes and 0 means No)
Homephone_Tag	Homephone Number provided by Client (1 means Yes and 0 means No)
Workphone_Working	Was workphone number reachable (1 means Yes and 0 means No)
Client_Occupation	Client Occupation type
Client_Family_Members	Number of family members does client have
Cleint_City_Rating	Client city rating. 3 denotes best and 2 denotes good and 1 denotes average
Application_Process_Day	Day of the week on which client applied for the loan (0-Sun, 1-Mon, 2-Tues, 3-Wed, 4-Thrus, 5-Fri, 6-Sat)
Application_Process_Hour	hour of the day on which client applied for the loan
Client_Permanent_Match_Tag	Indication if client contact address does not match permanent address.
Client_Contact_Work_Tag	Indication if client work address does not match contact address.
Type_Organization	Type of organization where client works
Score_Source_1	Score sourced from other source. This is a normalzied socre
Score_Source_2	Score sourced from other source. This is a normalzied socre
Score_Source_3	Score sourced from other source. This is a normalzied socre
Social_Circle_Default	How many friends/family member of client defaulted on any loan payment in last 60 days
Phone_Change	How many days before the loan application, client changed his/her phone
Credit_Bureau	Total number of enquiries in last year
Default	1 means the client defaulted on loan payments and 0 means otherwise

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Warning message in lapply(data[, to_num], as.numeric):
"NA's introduced by coercion"
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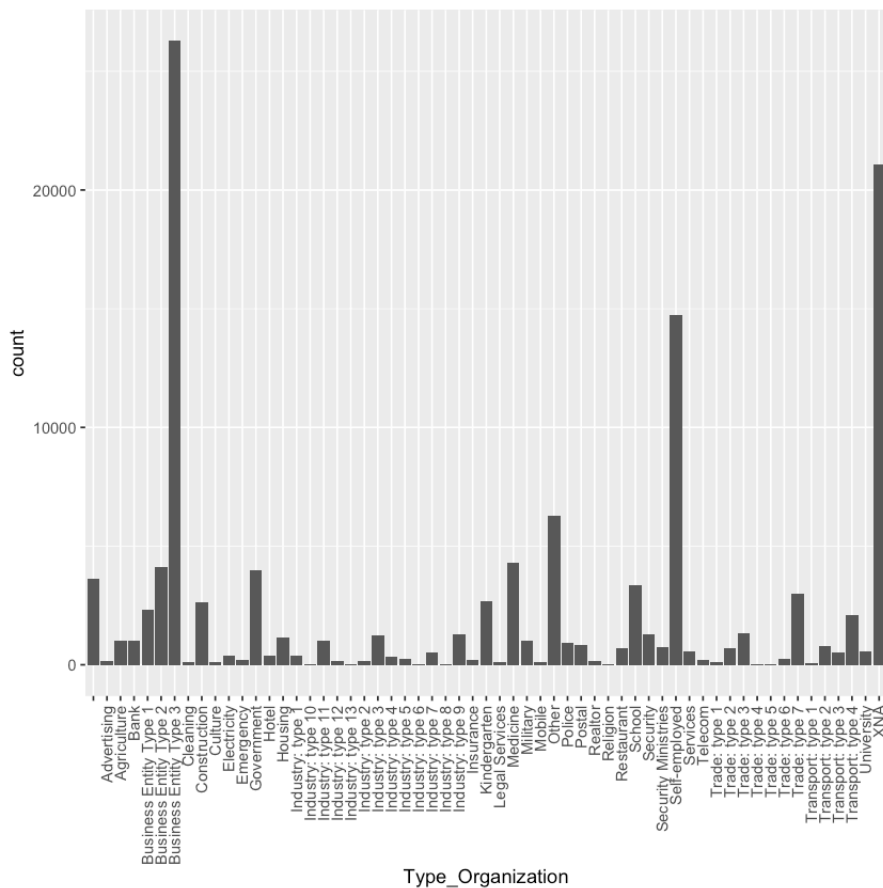
'data.frame': 121856 obs. of 40 variables:
 $ ID : int 12142509 12138936 12181264 12188929 1213
3385 12191614 12128086 12215264 12159147 12130547 ...
 $ Client_Income : num 6750 20250 18000 15750 33750 ...
 $ Car_Owned : int 0 1 0 0 1 0 1 0 1 0 ...
 $ Bike_Owned : int 0 0 0 0 0 1 1 0 1 0 ...
 $ Active_Loan : int 1 1 1 1 1 1 0 1 0 0 ...
 $ House_Own : int 0 NA 0 1 0 1 1 1 1 1 ...
 $ Child_Count : int 0 0 1 0 2 1 0 0 1 0 ...
 $ Credit_Amount : num 61191 15282 59527 53870 133988 ...
 $ Loan_Annuity : num 3417 1827 2788 2295 3547 ...
 $ Accompany_Client : Factor w/ 8 levels "", "##", "Alone", ...: 3 3 3
3 3 3 3 3 8 3 ...
 $ Client_Income_Type : Factor w/ 9 levels "", "Businessman", ...: 3 7 7
6 3 7 6 6 3 6 ...
 $ Client_Education : Factor w/ 6 levels "", "Graduation", ...: 6 2 3
6 6 6 6 6 2 6 ...
 $ Client_Marital_Status : Factor w/ 5 levels "", "D", "M", "S", ...: 3 3 5 3
3 5 4 3 3 5 ...
 $ Client_Gender : Factor w/ 4 levels "", "Female", "Male", ...: 3 3
3 3 2 2 3 3 2 3 ...
 $ Loan_Contract_Type : Factor w/ 3 levels "", "CL", "RL": 2 2 2 2 2 2
2 2 2 2 ...
 $ Client_Housing_Type : Factor w/ 7 levels "", "Family", "Home", ...: 3 3
2 3 3 3 3 3 3 3 ...
 $ Population_Region_Relative: num 0.02866 0.00857 0.0228 0.01056 0.02071 .
..
 $ Age_Days : num 13957 14162 16790 23195 11366 ...
 $ Employed_Days : num 1062 4129 5102 365243 2977 ...
 $ Registration_Days : num 6123 7833 NA NA 5516 ...
 $ ID_Days : num 383 21 331 775 4043 ...
 $ Own_House_Age : int NA 0 NA NA 6 NA 10 NA 14 NA ...
 $ Mobile_Tag : num 1 1 1 1 1 1 1 1 1 1 ...
 $ Homephone_Tag : int 1 0 0 0 0 0 0 0 0 0 ...

```

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$ Workphone_Working      : int    0 1 0 0 0 0 0 1 1 0 ...
$ Client_Occupation      : Factor w/ 19 levels "", "Accountants",...: 16 1
15 1 10 10 1 1 16 1 ...
$ Client_Family_Members  : int    2 2 2 2 4 2 1 2 3 1 ...
$ Cleint_City_Rating     : int    2 2 2 3 1 2 2 2 2 2 ...
$ Application_Process_Day : Factor w/ 7 levels "0","1","2","3",...: 7 4 5
3 4 3 4 5 5 NA ...
$ Application_Process_Hour : Factor w/ 26 levels "", "#NA","0","1",...: 12 5
1 10 1 5 9 10 8 26 ...
$ Client_Permanent_Match_Tag: Factor w/ 2 levels "No","Yes": 2 2 2 2 2 2 2
2 2 2 ...
$ Client_Contact_Work_Tag : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 2 2
2 2 2 ...
$ Type_Organization      : Factor w/ 59 levels "", "Advertising",...: 44 1
3 44 59 7 35 59 59 44 59 ...
$ Score_Source_1         : num    0.568 0.563 NA NA 0.508 ...
$ Score_Source_2         : num    0.479 0.215 0.553 0.135 0.301 ...
$ Score_Source_3         : num    NA NA 0.33 0.631 0.356 ...
$ Social_Circle_Default  : num    0.0186 NA 0.0742 NA 0.2021 ...
$ Phone_Change           : int    63 NA 277 1700 674 739 0 1687 1611 533 .
..
$ Credit_Bureau          : int    NA NA 0 3 1 0 3 4 0 5 ...
$ Default                : int    0 0 0 0 0 0 0 0 0 0 ...

```



(Intercept)	-2.4326990	Type_OrganizationAgriculture	0.1687661
Type_OrganizationBank	-0.1098251	Type_OrganizationConstruction	0.3364007
Type_OrganizationInsurance	-0.1514028	Type_OrganizationPolice	-0.2290746
Type_OrganizationSchool	-0.2566446	Type_OrganizationSecurity	0.1275355
Type_OrganizationSecurity Ministries	-0.1755129	Type_OrganizationSelf-employed	0.1933645
Type_OrganizationTransport: type 3	0.4115849	Type_OrganizationXNA	-0.3215006

'Other' · 'Agriculture' · 'Bank' · 'Construction' · 'Insurance' · 'Police' · 'School' · 'Security' · 'Security Ministries' · 'Self-employed' · 'Transport: type 3' · 'XNA'

(Intercept)	-2.5757229	Client_OccupationAccountants	-0.1022047
Client_OccupationCleaning	0.1137728	Client_OccupationCooking	0.3599217
Client_OccupationDrivers	0.4119356	Client_OccupationLaborers	0.3843504
Client_OccupationLow-skill Laborers	0.8294368	Client_OccupationSales	0.2870227
Client_OccupationSecurity	0.3847612		

'Other' · 'Accountants' · 'Cleaning' · 'Cooking' · 'Drivers' · 'Laborers' · 'Low-skill Laborers' · 'Sales' · 'Security'

Client_Income	2	Car_Owned	3
Bike_Owned	4	Active_Loan	5
House_Own	6	Child_Count	7
Credit_Amount	8	Loan_Annuity	9
Population_Region_Relative	17	Age_Days	18
Employed_Days	19	Registration_Days	20
ID_Days	21	Own_House_Age	22
Client_Family_Members	27	Cleint_City_Rating	28
Application_Process_Day	29	Score_Source_1	34
Score_Source_2		Score_Source_3	

35	36				
Social_Circle_Default	Phone_Change				
37	38				
Credit_Bureau					
39					
ID	Client_Income				
0	3622				
Car_Owned	Bike_Owned				
3581	3624				
Active_Loan	House_Own				
3635	3661				
Child_Count	Credit_Amount				
3638	3637				
Loan_Annuity	Accompany_Client				
4826	0				
Client_Income_Type	Client_Education				
0	0				
Client_Marital_Status	Client_Gender				
0	0				
Loan_Contract_Type	Client_Housing_Type				
0	0				
Population_Region_Relative	Age_Days				
4868	3617				
Employed_Days	Registration_Days				
3666	3631				
ID_Days	Own_House_Age				
5985	80095				
Mobile_Tag	Homephone_Tag				
0	0				
Workphone_Working	Client_Occupation				
0	0				
Client_Family_Members	Cleint_City_Rating				
2410	2409				
Application_Process_Day	Application_Process_Hour				
2428	0				
Client_Permanent_Match_Tag	Client_Contact_Work_Tag				
0	0				
Type_Organization	Score_Source_1				
0	68835				
Score_Source_2	Score_Source_3				
5686	26922				
Social_Circle_Default	Phone_Change				
61928	3664				
Credit_Bureau	Default				
18540	0				
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.08079	0.00000	1.00000

'data.frame': 121856 obs. of 63 variables:

```
$ ID : int 12142509 12138936 12181264 12188929 1
2133385 12191614 12128086 12215264 12159147 12130547 ...
$ Client_Income : num 6750 20250 18000 15750 33750 ...
$ Car_Owned : int 0 1 0 0 1 0 1 0 1 0 ...
```

```

$ Bike_Owned           : int  0 0 0 0 0 1 1 0 1 0 ...
$ Active_Loan          : int  1 1 1 1 1 1 0 1 0 0 ...
$ House_Own            : int  0 0 0 1 0 1 1 1 1 1 ...
$ Child_Count          : int  0 0 1 0 2 1 0 0 1 0 ...
$ Credit_Amount        : num  61191 15282 59527 53870 133988 ...
$ Loan_Annuity         : num   3417 1827 2788 2295 3547 ...
$ Accompany_Client     : Factor w/ 7 levels "Nothing","Alone",...: 2
2 2 2 2 2 2 2 7 2 ...
$ Client_Income_Type   : Factor w/ 9 levels "", "Businessman",...: 3
7 7 6 3 7 6 6 3 6 ...
$ Client_Education     : Factor w/ 6 levels "", "Graduation",...: 6 2
3 6 6 6 6 6 2 6 ...
$ Client_Marital_Status : Factor w/ 5 levels "", "D", "M", "S",...: 3 3
5 3 3 5 4 3 3 5 ...
$ Client_Gender        : Factor w/ 4 levels "", "Female", "Male",...:
3 3 3 3 2 2 3 3 2 3 ...
$ Loan_Contract_Type   : Factor w/ 3 levels "", "CL", "RL": 2 2 2 2 2
2 2 2 2 2 ...
$ Client_Housing_Type  : Factor w/ 7 levels "", "Family", "Home",...:
3 3 2 3 3 3 3 3 3 3 ...
$ Population_Region_Relative : num  0.02866 0.00857 0.0228 0.01056 0.0207
1 ...
$ Age_Days             : num  13957 14162 16790 23195 11366 ...
$ Employed_Days        : num  1062 4129 5102 365243 2977 ...
$ Registration_Days    : num  6123 7833 0 0 5516 ...
$ ID_Days              : num  383 21 331 775 4043 ...
$ Own_House_Age        : int  0 0 0 0 6 0 10 0 14 0 ...
$ Mobile_Tag           : num  1 1 1 1 1 1 1 1 1 1 ...
$ Homephone_Tag        : int  1 0 0 0 0 0 0 0 0 0 ...
$ Workphone_Working    : int  0 1 0 0 0 0 0 1 1 0 ...
$ Client_Occupation    : Factor w/ 9 levels "Other","Accountants",.
.: 8 1 1 1 6 6 1 1 8 1 ...
$ Client_Family_Members : int  2 2 2 2 4 2 1 2 3 1 ...
$ Cleint_City_Rating    : int  2 2 2 3 1 2 2 2 2 2 ...
$ Application_Process_Day : Factor w/ 7 levels "0","1","2","3",...: 7 4
5 3 4 3 4 5 5 1 ...
$ Application_Process_Hour : Factor w/ 8 levels "Other","0","1",...: 4 1
1 1 1 1 1 1 1 1 ...
$ Client_Permanent_Match_Tag : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 2
2 2 2 2 ...
$ Client_Contact_Work_Tag : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 2
2 2 2 2 ...
$ Type_Organization    : Factor w/ 12 levels "Other","Agriculture",
.: 10 1 10 12 1 1 12 12 10 12 ...
$ Score_Source_1       : num  0.568 0.563 0 0 0.508 ...
$ Score_Source_2       : num  0.479 0.215 0.553 0.135 0.301 ...
$ Score_Source_3       : num  0 0 0.33 0.631 0.356 ...
$ Social_Circle_Default : num  0.0186 0 0.0742 0 0.2021 ...
$ Phone_Change         : int  63 0 277 1700 674 739 0 1687 1611 533
...
$ Credit_Bureau        : int  0 0 0 3 1 0 3 4 0 5 ...
$ Default              : int  0 0 0 0 0 0 0 0 0 0 ...
$ Client_Income_na     : num  0 0 0 0 0 0 0 0 0 0 ...

```

```

$ Car_Owned_na      : num  0 0 0 0 0 0 0 0 0 0 0 ...
$ Bike_Owned_na     : num  0 0 0 0 0 0 0 0 0 0 0 ...
$ Active_Loan_na    : num  0 0 0 0 0 0 0 0 0 0 0 ...
$ House_Own_na      : num  0 1 0 0 0 0 0 0 0 0 0 ...
$ Child_Count_na    : num  0 0 0 0 0 0 0 0 0 0 0 ...
$ Credit_Amount_na  : num  0 0 0 0 0 0 0 0 0 0 0 ...
$ Loan_Annuity_na   : num  0 0 0 0 0 0 0 0 0 0 0 ...
$ Population_Region_Relative_na: num  0 0 0 0 0 0 0 0 0 0 0 ...
$ Age_Days_na       : num  0 0 0 0 0 0 0 0 0 1 0 ...
$ Employed_Days_na  : num  0 0 0 0 0 0 0 0 0 0 0 ...
$ Registration_Days_na: num  0 0 1 1 0 0 0 0 0 0 0 ...
$ ID_Days_na        : num  0 0 0 0 0 0 0 0 0 0 0 ...
$ Own_House_Age_na  : num  1 0 1 1 0 1 0 1 0 1 1 ...
$ Client_Family_Members_na: num  0 0 0 0 0 0 0 0 0 0 0 ...
$ Cleint_City_Rating_na: num  0 0 0 0 0 0 0 0 0 0 0 ...
$ Application_Process_Day_na: num  0 0 0 0 0 0 0 0 0 0 1 ...
$ Score_Source_1_na : num  0 0 1 1 0 1 0 0 0 0 0 ...
$ Score_Source_2_na : num  0 0 0 0 0 0 0 0 0 0 0 ...
$ Score_Source_3_na : num  1 1 0 0 0 0 0 0 0 0 0 ...
$ Social_Circle_Default_na: num  0 1 0 1 0 0 0 1 0 1 1 ...
$ Phone_Change_na   : num  0 1 0 0 0 0 0 0 0 0 0 ...
$ Credit_Bureau_na  : num  1 1 0 0 0 0 0 0 0 0 0 ...

```

```
named integer(0)
```

Warning message:

"glm.fit: fitted probabilities numerically 0 or 1 occurred"

Call:

```
glm(formula = Default ~ ., family = "binomial", data = training)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.4731	-0.4395	-0.3256	-0.2335	8.4904

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.105e+01	3.247e+02	-0.034	0.972852
Client_Income	-3.237e-06	1.531e-06	-2.115	0.034467
*				
Car_Owned	-6.663e-03	1.403e-01	-0.047	0.962136
Bike_Owned	-2.560e-03	2.348e-02	-0.109	0.913172
Active_Loan	1.008e-02	2.210e-02	0.456	0.648235
House_Own	8.779e-02	2.519e-02	3.485	0.000491

Child_Count	-2.722e-02	5.153e-02	-0.528	0.597305
Credit_Amount	-1.222e-06	4.456e-07	-2.742	0.006114
**				
Loan_Annuity	7.187e-05	1.247e-05	5.763	8.28e-09

Accompany_ClientAlone	1.252e-01	9.661e-02	1.296	0.194935
Accompany_ClientGroup	-1.868e-01	4.150e-01	-0.450	0.652576

Accompany_ClientKids	8.896e-03	1.479e-01	0.060	0.952055
Accompany_ClientOthers	9.634e-02	1.519e-01	0.634	0.526009
Accompany_ClientPartner	4.681e-02	1.118e-01	0.419	0.675540
Accompany_ClientRelative	7.231e-02	1.009e-01	0.716	0.473801
Client_Income_TypeBusinessman	-1.072e+01	3.247e+02	-0.033	0.973656
Client_Income_TypeCommercial	-4.835e-02	6.965e-02	-0.694	0.487537
Client_Income_TypeGovt Job	-1.216e-01	8.239e-02	-1.476	0.140043
Client_Income_TypeMaternity leave	-8.435e+00	2.280e+02	-0.037	0.970494
Client_Income_TypeRetired	4.694e-02	1.466e-01	0.320	0.748862
Client_Income_TypeService	1.729e-02	6.698e-02	0.258	0.796335
Client_Income_TypeStudent	-9.687e+00	1.110e+02	-0.087	0.930427
Client_Income_TypeUnemployed	2.345e+00	8.620e-01	2.721	0.006517
**				
Client_EducationGraduation	-1.775e-01	7.148e-02	-2.483	0.013020
*				
Client_EducationGraduation dropout	-3.075e-02	8.887e-02	-0.346	0.729386
Client_EducationJunior secondary	2.675e-01	1.105e-01	2.421	0.015458
*				
Client_EducationPost Grad	-1.379e+00	1.026e+00	-1.344	0.178956
Client_EducationSecondary	1.958e-01	6.653e-02	2.943	0.003246
**				
Client_Marital_StatusD	1.353e-01	8.723e-02	1.551	0.120821
Client_Marital_StatusM	-1.461e-03	6.773e-02	-0.022	0.982791
Client_Marital_StatusS	1.300e-01	8.013e-02	1.622	0.104805
Client_Marital_StatusW	1.492e-01	9.347e-02	1.596	0.110468
Client_GenderFemale	1.535e-01	7.964e-02	1.928	0.053912
.				
Client_GenderMale	-1.166e-01	7.863e-02	-1.483	0.138144
Client_GenderXNA	-1.015e+01	1.821e+02	-0.056	0.955539
Loan_Contract_TypeCL	1.736e-02	6.312e-02	0.275	0.783251
Loan_Contract_TypeRL	-3.469e-01	7.579e-02	-4.577	4.72e-06

Client_Housing_TypeFamily	-3.316e-02	7.627e-02	-0.435	0.663711
Client_Housing_TypeHome	-4.760e-02	6.255e-02	-0.761	0.446679
Client_Housing_TypeMunicipal	1.691e-01	8.364e-02	2.021	0.043230
*				
Client_Housing_TypeOffice	-2.047e-01	1.431e-01	-1.430	0.152650
Client_Housing_TypeRental	3.353e-02	9.741e-02	0.344	0.730709
Client_Housing_TypeShared	1.994e-01	1.762e-01	1.131	0.257851
Population_Region_Relative	-1.858e-02	7.789e-02	-0.239	0.811428
Age_Days	-1.101e-05	3.853e-06	-2.857	0.004276
**				
Employed_Days	-7.483e-07	4.073e-07	-1.837	0.066173
.				
Registration_Days	-1.020e-05	3.519e-06	-2.899	0.003743
**				
ID_Days	-4.952e-05	7.868e-06	-6.294	3.10e-10

Own_House_Age	8.511e-03	1.456e-03	5.846	5.03e-09

Mobile_Tag	1.043e+01	3.247e+02	0.032	0.974376
Homephone_Tag	5.972e-02	2.869e-02	2.081	0.037406
*				

Workphone_Working	-9.256e-02	2.690e-02	-3.440	0.000581

Client_OccupationAccountants	4.091e-02	7.723e-02	0.530	0.596331
Client_OccupationCleaning	1.288e-01	8.825e-02	1.460	0.144349
Client_OccupationCooking	2.120e-01	7.282e-02	2.911	0.003606
**				
Client_OccupationDrivers	1.544e-01	4.789e-02	3.223	0.001268
**				
Client_OccupationLaborers	1.174e-01	3.188e-02	3.682	0.000232

Client_OccupationLow-skill Laborers	3.338e-01	1.020e-01	3.273	0.001063
**				
Client_OccupationSales	1.174e-01	3.878e-02	3.026	0.002476
**				
Client_OccupationSecurity	2.335e-01	7.816e-02	2.988	0.002809
**				
Client_Family_Members	7.137e-02	5.065e-02	1.409	0.158801
Cleint_City_Rating	1.676e-01	2.410e-02	6.953	3.57e-12

Application_Process_Day1	4.747e-02	5.677e-02	0.836	0.403040
Application_Process_Day2	1.332e-01	5.598e-02	2.380	0.017329
*				
Application_Process_Day3	1.614e-01	5.622e-02	2.871	0.004090
**				
Application_Process_Day4	9.431e-02	5.651e-02	1.669	0.095127
.				
Application_Process_Day5	1.195e-01	5.648e-02	2.115	0.034408
*				
Application_Process_Day6	1.259e-01	5.939e-02	2.119	0.034050
*				
Application_Process_Hour0	1.195e+00	5.167e-01	2.313	0.020722
*				
Application_Process_Hour1	-1.036e+01	5.961e+01	-0.174	0.862049
Application_Process_Hour17	-1.712e-01	5.651e-02	-3.030	0.002446
**				
Application_Process_Hour19	-2.230e-01	1.191e-01	-1.872	0.061217
.				
Application_Process_Hour5	3.023e-02	9.070e-02	0.333	0.738917
Application_Process_Hour6	2.175e-02	7.394e-02	0.294	0.768647
Application_Process_Hour7	9.449e-02	5.955e-02	1.587	0.112575
Client_Permanent_Match_TagYes	-1.350e-01	3.710e-02	-3.640	0.000273

Client_Contact_Work_TagYes	2.344e-02	2.832e-02	0.828	0.407722
Type_OrganizationAgriculture	1.309e-01	1.033e-01	1.268	0.204895
Type_OrganizationBank	-3.833e-01	1.476e-01	-2.596	0.009430
**				
Type_OrganizationConstruction	2.966e-01	6.451e-02	4.597	4.29e-06

Type_OrganizationInsurance	-9.223e-01	4.564e-01	-2.021	0.043298
*				
Type_OrganizationPolice	-5.639e-01	1.678e-01	-3.360	0.000781

Type_OrganizationSchool	-3.059e-01	8.234e-02	-3.715	0.000203

```

***
Type_OrganizationSecurity          5.412e-02  1.103e-01  0.491 0.623623
Type_OrganizationSecurity Ministries -5.009e-01  1.836e-01  -2.728 0.006363
**
Type_OrganizationSelf-employed      1.315e-01  3.284e-02  4.003 6.24e-05
***
Type_OrganizationTransport: type 3   4.612e-01  1.364e-01  3.380 0.000724
***
Type_OrganizationXNA                 6.711e-02  1.497e-01  0.448 0.653879
Score_Source_1                      -1.334e+00  9.281e-02 -14.369 < 2e-16
***
Score_Source_2                      -1.579e+00  5.775e-02 -27.346 < 2e-16
***
Score_Source_3                      -2.843e+00  6.459e-02 -44.015 < 2e-16
***
Social_Circle_Default               -6.578e-01  1.751e-01  -3.756 0.000173
***
Phone_Change                        -8.190e-05  1.511e-05  -5.420 5.97e-08
***
Credit_Bureau                      1.277e-02  6.359e-03  2.008 0.044607
*
Client_Income_na                    -7.235e-02  6.971e-02  -1.038 0.299275
Car_Owned_na                       -1.455e-02  7.832e-02  -0.186 0.852611
Bike_Owned_na                      1.498e-01  6.117e-02  2.449 0.014319
*
Active_Loan_na                     6.138e-03  6.439e-02  0.095 0.924067
House_Own_na                       1.327e-01  6.457e-02  2.055 0.039870
*
Child_Count_na                     -7.001e-03  6.839e-02  -0.102 0.918470
Credit_Amount_na                  -1.484e-01  7.066e-02  -2.100 0.035747
*
Loan_Annuity_na                    2.436e-01  6.368e-02  3.825 0.000131
***
Population_Region_Relative_na       8.365e-02  5.374e-02  1.557 0.119576
Age_Days_na                       -1.611e-01  8.691e-02  -1.853 0.063831
.
Employed_Days_na                   -1.221e-01  6.948e-02  -1.758 0.078776
.
Registration_Days_na                -1.378e-01  6.800e-02  -2.026 0.042782
*
ID_Days_na                         -1.247e-01  5.481e-02  -2.274 0.022945
*
Own_House_Age_na                   3.784e-01  1.400e-01  2.703 0.006873
**
Client_Family_Members_na            1.697e-01  1.361e-01  1.247 0.212510
Cleint_City_Rating_na              3.718e-01  9.258e-02  4.016 5.91e-05
***
Application_Process_Day_na           1.682e-01  9.069e-02  1.855 0.063600
.
Score_Source_1_na                   -3.967e-01  4.576e-02  -8.669 < 2e-16
***
Score_Source_2_na                   -7.550e-01  5.812e-02 -12.991 < 2e-16
***

```

Score_Source_3_na	-1.439e+00	4.558e-02	-31.584	< 2e-16

Social_Circle_Default_na	2.334e-02	2.941e-02	0.793	0.427556
Phone_Change_na	2.725e-02	6.270e-02	0.435	0.663797
Credit_Bureau_na	2.394e-01	4.278e-02	5.596	2.19e-08

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 68410 on 121855 degrees of freedom
 Residual deviance: 61568 on 121739 degrees of freedom
 AIC: 61802

Number of Fisher Scoring iterations: 11

117

1 2 3 4 5
 No No No No No
 Levels: No Yes
 'No' · 'Yes'

'No' · 'Yes'

Confusion Matrix and Statistics

	Reference	
Prediction	No	Yes
No	99889	6173
Yes	12122	3672

Accuracy : 0.8499
 95% CI : (0.8478, 0.8519)
 No Information Rate : 0.9192
 P-Value [Acc > NIR] : 1

Kappa : 0.2076

Mcnemar's Test P-Value : <2e-16

Sensitivity : 0.37298
 Specificity : 0.89178
 Pos Pred Value : 0.23249
 Neg Pred Value : 0.94180
 Prevalence : 0.08079
 Detection Rate : 0.03013
 Detection Prevalence : 0.12961
 Balanced Accuracy : 0.63238

'Positive' Class : Yes

Setting levels: control = No, case = Yes

Setting direction: controls < cases

